

MAKHACHKALA MEZHDUNARODNNY, Russian Federation

WMO: 377352

Lat: 42.817N

Lon: 47.652E

Elev: 4

StdP: 101.28

Time Zone: 3.00 (E03)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-7.1	-11.2	1.4	-7.3	-9.2	1.7	-5.1	15.0	3.6	13.3	2.6	4.5	340	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	33.1	22.6	32.0	22.4	30.9	22.5	25.2	29.7	24.5	29.3	23.8	28.8	5.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	18.8	28.3	23.0	17.8	27.6	22.1	16.7	27.1	76.9	29.7	74.0	28.8	70.9	28.8	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.2	11.2	9.7	-13.1	35.9	4.7	0.7	-16.5	36.5	-19.3	36.9	-21.9	37.3	-25.3	37.8		
			-11.9	26.5	3.0	1.6	-14.1	27.7	-15.8	28.6	-17.5	29.5	-19.6	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.3	1.6	1.8	5.6	10.7	17.5	23.0	25.7	25.4	20.7	14.4	8.0	4.3	(6)
		DBStd	9.30	4.10	4.26	2.46	3.37	3.09	2.73	2.20	2.43	3.03	3.52	3.77	3.51	
		HDD10.0	918	261	228	137	29	0	0	0	0	8	78	177		
		HDD18.3	2543	519	462	394	230	53	2	0	12	126	311	434		
		CDD10.0	2123	0	0	2	49	232	390	487	477	322	146	17	1	
		CDD18.3	707	0	0	0	1	27	142	229	219	84	6	0	0	
		CDH23.3	6482	0	0	0	8	183	1239	2364	2194	483	10	0	0	
		CDH26.7	2229	0	0	0	0	36	377	897	833	87	0	0	0	
(14)	Wind	WSAvg	4.4	4.4	4.4	4.8	4.5	4.1	4.2	4.3	4.2	4.5	4.5	4.5	4.6	(14)
(15)	Precipitation	PrecAvg	384	25	27	21	20	39	36	29	30	45	46	36	29	(15)
(16)		PrecMax	544	77	105	49	34	67	78	66	87	165	126	108	105	(16)
(17)		PrecMin	280	6	7	2	4	4	13	11	0	0	7	0	1	(17)
(18)		PrecStd	75	20	23	13	8	19	16	14	22	43	32	33	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	11.0	11.1	16.1	24.2	30.0	33.2	35.1	35.0	31.0	24.8	18.2	13.2		(19)
(20)		0.4% MCWB	6.5	7.6	10.8	16.1	19.1	21.7	23.2	22.7	21.8	18.4	14.9	9.6		(20)
(21)		2% DB	8.8	8.9	13.0	21.1	27.2	31.8	33.1	33.1	28.9	22.8	16.0	11.2		(21)
(22)		2% MCWB	6.0	6.6	9.0	14.8	19.0	21.5	22.8	22.8	22.0	18.1	13.5	8.9		(22)
(23)		5% DB	7.1	7.2	11.1	18.2	25.1	30.1	32.0	32.0	27.2	21.1	14.8	10.1		(23)
(24)		5% MCWB	5.2	5.6	7.8	13.1	18.4	21.5	22.6	22.7	21.0	17.3	12.4	8.3		(24)
(25)		10% DB	6.1	6.2	9.8	16.2	23.2	28.9	30.9	30.9	26.0	20.0	13.1	9.1		(25)
(26)		10% MCWB	4.7	5.1	7.2	12.5	17.4	20.9	22.6	23.0	20.6	16.7	11.2	7.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	7.9	8.2	11.2	16.8	20.8	25.0	26.1	26.3	24.9	19.9	15.5	11.1		(27)
(28)		0.4% MCDB	9.8	10.4	15.1	23.2	26.1	29.7	30.5	30.2	27.6	23.0	17.5	12.5		(28)
(29)		2% WB	6.3	6.7	9.5	15.1	19.7	23.7	25.1	25.4	23.1	18.7	13.8	9.7		(29)
(30)		2% MCDB	7.5	7.8	12.6	19.9	25.2	28.7	29.9	29.7	26.9	21.6	15.7	10.5		(30)
(31)		5% WB	5.5	5.9	8.2	13.9	18.9	22.7	24.4	24.6	22.1	17.8	12.6	8.7		(31)
(32)		5% MCDB	6.5	6.8	10.3	17.8	24.3	28.0	29.5	29.3	26.1	20.5	14.5	9.8		(32)
(33)		10% WB	4.8	5.2	7.3	12.7	18.1	21.9	23.7	23.9	21.0	17.0	11.6	7.7		(33)
(34)		10% MCDB	5.7	6.0	8.8	16.0	22.7	27.2	28.9	28.9	24.9	19.6	13.2	8.7		(34)
(35)	Mean Daily Temperature Range	MDBR	5.8	5.6	6.5	8.0	9.2	9.6	9.0	8.9	8.1	7.1	6.5	5.7		(35)
(36)		5% DB MCDBR	8.0	7.0	9.5	10.7	12.2	11.3	10.6	10.5	9.1	8.4	8.8	7.0		(36)
(37)		5% WB MCWBR	6.0	5.2	6.2	6.3	5.9	5.5	5.2	5.6	5.2	5.4	6.4	5.2		(37)
(38)		5% WB MCDBR	6.8	6.2	8.2	10.1	10.7	9.9	9.5	9.5	8.8	7.6	7.8	5.7		(38)
(39)		5% WB MCWBR	5.3	4.8	5.9	6.2	5.5	5.6	5.3	5.9	5.5	5.2	5.9	4.6		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.364	0.401	0.441	0.504	0.478	0.478	0.512	0.511	0.475	0.421	0.380	0.369		(40)
(41)		taud	2.236	2.130	2.075	1.960	2.107	2.165	2.089	2.083	2.140	2.238	2.273	2.231		(41)
(42)		Ebn at Noon	742	772	789	769	805	805	771	755	750	745	719	693		(42)
(43)		Edh at Noon	93	122	145	175	156	147	157	154	135	108	90	87		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.45	2.24	3.25	4.55	5.55	6.16	6.07	5.44	4.18	2.67	1.75	1.30		(44)
(45)		RadStd	0.17	0.30	0.29	0.47	0.48	0.36	0.45	0.43	0.40	0.33	0.18	0.16		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page